Before/after analysis — collect actual data

No learner outcomes have been measured for this package. The blank fields below are intentional.

Method

Take the 15-question quiz unaided before starting and after day 28. Use the same answer key, record testing conditions, and acknowledge familiarity with repeated questions. Also repeat two coding tasks with new values and ask the learner to explain each result. Obtain consent if another student participates; use a pseudonym, not identifying records.

Measure	Before	After	Change
Quiz score /15	Not collected	Not collected	Not calculated
Independent exercises correct	Not collected	Not collected	Not calculated
Hints required on matched tasks	Not collected	Not collected	Not calculated
Explanation quality /3	Not collected	Not collected	Not calculated

Explanation rubric: 0 cannot explain; 1 names operations; 2 traces values; 3 explains values and a boundary case. Quiz percentage-point change = (after − before) / 15 × 100. Report both raw scores. Do not calculate percentage growth when the baseline is zero.

Reflection to complete

Which misconception changed? What evidence shows independent understanding? Which tasks remain difficult? Did AI provide too much help? What will you revise? Do not describe a single learner comparison as proof that AI caused improvement.